Paintbrush

8.1、Introduction

MediaPipe is a data stream processing machine learning application development framework developed and open-sourced by Google. It is a graph-based data processing pipeline for building with various forms of data sources such as video, audio, sensor data, and any time series data. MediaPipe is cross-platform and can run on embedded platforms (Raspberry Pi, etc.), mobile devices (iOS and Android), workstations and servers, and supports mobile GPU acceleration. MediaPipe provides cross-platform, customizable ML solutions for real-time and streaming media.

MediaPipe's core framework is implemented in C++ and provides support for Java and Objective C. The main concepts of MediaPipe include Packet, Stream, Calculator, Graph, and Subgraph.

MediaPipe features:

- End-to-end acceleration: Built-in fast ML inference and processing accelerates even on ordinary hardware.
- Build once, deploy anywhere: Unified solutions work on Android, iOS, desktop/cloud, web, and IoT.
- Ready-to-use solutions: Cutting-edge ML solutions that showcase all framework features.
- Free and open source: Framework and solutions under Apache 2.0, fully extensible and customizable.

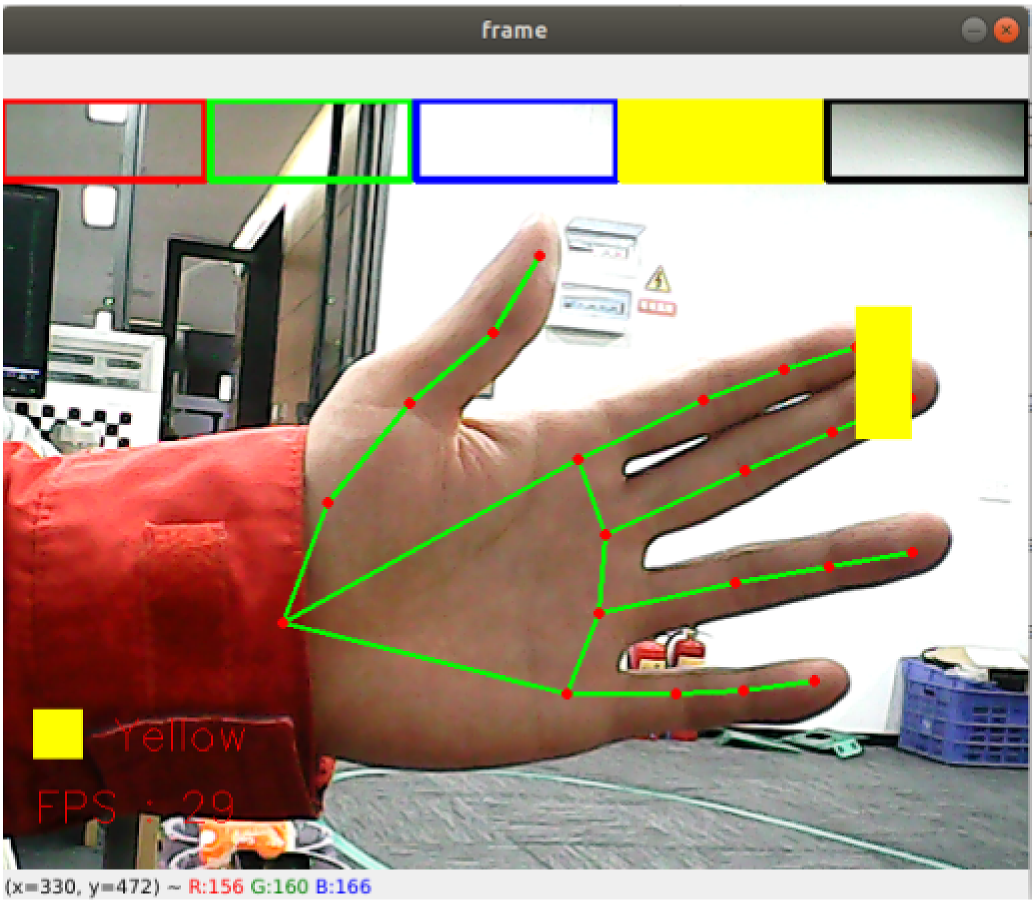
8.2、Paintbrush

When the right hand index finger and middle finger are together, it is the selection state, and a color selection box pops up at the same time. When the two fingertips move to the corresponding color position, that color is selected (black is the eraser); when the index finger and middle finger are separated, it is the drawing state, and you can draw freely on the canvas.

8.2.1、Startup

Enter in terminal,

```
ros2 run dofbot_mediapipe 08_VirtualPaint
```



8.2.2、Source Code

Source location: /home/yahboom/dofbot_ws/src/dofbot_mediapipe/dofbot_mediapipe/08_VirtualPaint.py

```
#!/usr/bin/env python3
# encoding: utf-8

import math
import time
import cv2 as cv
import numpy as np
import mediapipe as mp
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge

class HandDetector:
    def __init__(self, mode=False, maxHands=2, detectorCon=0.5, trackCon=0.5):
        self.tipIds = [4, 8, 12, 16, 20]
        self.mpHand = mp.solutions.hands
        self.mpDraw = mp.solutions.drawing_utils
        self.hands = self.mpHand.Hands(
            static_image_mode=mode,
            max_num_hands=maxHands,
            min_detection_confidence=detectorCon,
            min_tracking_confidence=trackCon )
        self.lmDrawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 0,
255), thickness=-1, circle_radius=15)
        self.drawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 255,
0), thickness=10, circle_radius=10)
```

```
    def findHands(self, frame, draw=True):
        self.lmList = []
        img_RGB = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
        self.results = self.hands.process(img_RGB)
        if self.results.multi_hand_landmarks:
            for handLms in self.results.multi_hand_landmarks:
                if draw: self.mpDraw.draw_landmarks(frame, handLms,
self.mpHand.HAND_CONNECTIONS, self.lmDrawSpec, self.drawSpec)
                else: self.mpDraw.draw_landmarks(frame, handLms,
self.mpHand.HAND_CONNECTIONS)
            for id, lm in
enumerate(self.results.multi_hand_landmarks[0].landmark):
                h, w, c = frame.shape
                cx, cy = int(lm.x * w), int(lm.y * h)
                self.lmList.append([id, cx, cy])
        return frame, self.lmList

    def fingersUp(self):
        fingers=[]
        if (self.calc_angle(self.tipIds[0], self.tipIds[0] - 1, self.tipIds[0] -
2) > 150.0) and (
                self.calc_angle(self.tipIds[0] - 1, self.tipIds[0] - 2,
self.tipIds[0] - 3) > 150.0): fingers.append(1)
        else:
            fingers.append(0)
        for id in range(1, 5):
            if self.lmList[self.tipIds[id]][2] < self.lmList[self.tipIds[id] -
2][2]:
                fingers.append(1)
            else:
                fingers.append(0)
        return fingers

    def get_dist(self, point1, point2):
        x1, y1 = point1
        x2, y2 = point2
        return abs(math.sqrt(math.pow(abs(y1 - y2), 2) + math.pow(abs(x1 - x2),
2)))

    def calc_angle(self, pt1, pt2, pt3):
        point1 = self.lmList[pt1][1], self.lmList[pt1][2]
        point2 = self.lmList[pt2][1], self.lmList[pt2][2]
        point3 = self.lmList[pt3][1], self.lmList[pt3][2]
        a = self.get_dist(point1, point2)
        b = self.get_dist(point2, point3)
        c = self.get_dist(point1, point3)
        try:
            radian = math.acos((math.pow(a, 2) + math.pow(b, 2) - math.pow(c,
2)) / (2 * a * b))
            angle = radian / math.pi * 180
        except:
            angle = 0
        return abs(angle)

class HandPaintingNode(Node):
    def __init__(self):
        super().__init__('hand_painting_node')
```

```
        self.publisher_ = self.create_publisher(Image, 'hand_painting_image',
10)
        self.timer = self.create_timer(0.1, self.timer_callback)
        self.bridge = CvBridge()
        self.hand_detector = HandDetector(detectorCon=0.85)
        self.xp = self.yp = self.pTime = self.boxx = 0
        self.tipIds = [4, 8, 12, 16, 20]
        self.imgCanvas = np.zeros((480, 640, 3), np.uint8)
        self.brushThickness = 5
        self.eraserThickness = 100
        self.top_height = 50
        self.Color = "Red"
        self.ColorList = {
            'Red': (0, 0, 255),
            'Green': (0, 255, 0),
            'Blue': (255, 0, 0),
            'Yellow': (0, 255, 255),
            'Black': (0, 0, 0),
        }
        self.capture = cv.VideoCapture(0, cv.CAP_V4L2)
        self.capture.set(6, cv.VideoWriter.fourcc('M', 'J', 'P', 'G'))
        self.capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
        self.capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)

    def timer_callback(self):
        ret, frame = self.capture.read()
        if not ret:
            self.get_logger().error('Failed to capture frame')
            return

        h, w, c = frame.shape
        frame, lmList = self.hand_detector.findHands(frame, draw=False)
        if len(lmList) != 0:
            x1, y1 = lmList[8][1:]
            x2, y2 = lmList[12][1:]
            fingers = self.hand_detector.fingersUp()
            if fingers[1] and fingers[2]:
                if y1 < self.top_height:
                    if 0 < x1 < int(w / 5) - 1:
                        self.boxx = 0
                        self.Color = "Red"
                    if int(w / 5) < x1 < int(w * 2 / 5) - 1:
                        self.boxx = int(w / 5)
                        self.Color = "Green"
                    elif int(w * 2 / 5) < x1 < int(w * 3 / 5) - 1:
                        self.boxx = int(w * 2 / 5)
                        self.Color = "Blue"
                    elif int(w * 3 / 5) < x1 < int(w * 4 / 5) - 1:
                        self.boxx = int(w * 3 / 5)
                        self.Color = "Yellow"
                    elif int(w * 4 / 5) < x1 < w - 1:
                        self.boxx = int(w * 4 / 5)
                        self.Color = "Black"
                cv.rectangle(frame, (x1, y1 - 25), (x2, y2 + 25),
self.ColorList[self.Color], cv.FILLED)
                cv.rectangle(frame, (self.boxx, 0), (self.boxx + int(w / 5),
self.top_height), self.ColorList[self.Color], cv.FILLED)
```

```
                cv.rectangle(frame, (0, 0), (int(w / 5) - 1, self.top_height), self.ColorList['Red'], 3)
                cv.rectangle(frame, (int(w / 5) + 2, 0), (int(w * 2 / 5) - 1, self.top_height), self.ColorList['Green'], 3)
                cv.rectangle(frame, (int(w * 2 / 5) + 2, 0), (int(w * 3 / 5) - 1, self.top_height), self.ColorList['Blue'], 3)
                cv.rectangle(frame, (int(w * 3 / 5) + 2, 0), (int(w * 4 / 5) - 1, self.top_height), self.ColorList['Yellow'], 3)
                cv.rectangle(frame, (int(w * 4 / 5) + 2, 0), (w - 1, self.top_height), self.ColorList['Black'], 3)
            if fingers[1] and fingers[2] == False and math.hypot(x2 - x1, y2 - y1) > 50:
                if self.xp == self.yp == 0: self.xp, self.yp = x1, y1
                if self.Color == 'Black':
                    cv.line(frame, (self.xp, self.yp), (x1, y1), self.ColorList[self.Color], self.eraserThickness)
                    cv.line(self.imgCanvas, (self.xp, self.yp), (x1, y1), self.ColorList[self.Color], self.eraserThickness)
                else:
                    cv.line(frame, (self.xp, self.yp), (x1, y1), self.ColorList[self.Color], self.brushThickness)
                    cv.line(self.imgCanvas, (self.xp, self.yp), (x1, y1), self.ColorList[self.Color], self.brushThickness)
                cv.circle(frame, (x1, y1), 15, self.ColorList[self.Color], cv.FILLED)
                self.xp, self.yp = x1, y1
            else: self.xp = self.yp = 0

        imgGray = cv.cvtColor(self.imgCanvas, cv.COLOR_BGR2GRAY)
        _, imgInv = cv.threshold(imgGray, 50, 255, cv.THRESH_BINARY_INV)
        imgInv = cv.cvtColor(imgInv, cv.COLOR_GRAY2BGR)
        frame = cv.bitwise_and(frame, imgInv)
        frame = cv.bitwise_or(frame, self.imgCanvas)

        if cv.waitKey(1) & 0xFF == ord('q'):
            self.capture.release()
            cv.destroyAllWindows()
            rclpy.shutdown()

        cTime = time.time()
        fps = 1 / (cTime - self.pTime)
        self.pTime = cTime
        text = "FPS : " + str(int(fps))
        cv.rectangle(frame, (20, h - 100), (50, h - 70), self.ColorList[self.Color], cv.FILLED)
        cv.putText(frame, self.Color, (70, h - 75), cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0, 255), 1)
        cv.putText(frame, text, (20, h-30), cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0, 255), 1)

        msg = self.bridge.cv2_to_imgmsg(frame, "bgr8")
        self.publisher_.publish(msg)

        cv.imshow('frame', frame)

def main(args=None):
    rclpy.init(args=args)
    hand_painting_node = HandPaintingNode()
```

```
    rclpy.spin(hand_painting_node)
    hand_painting_node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```